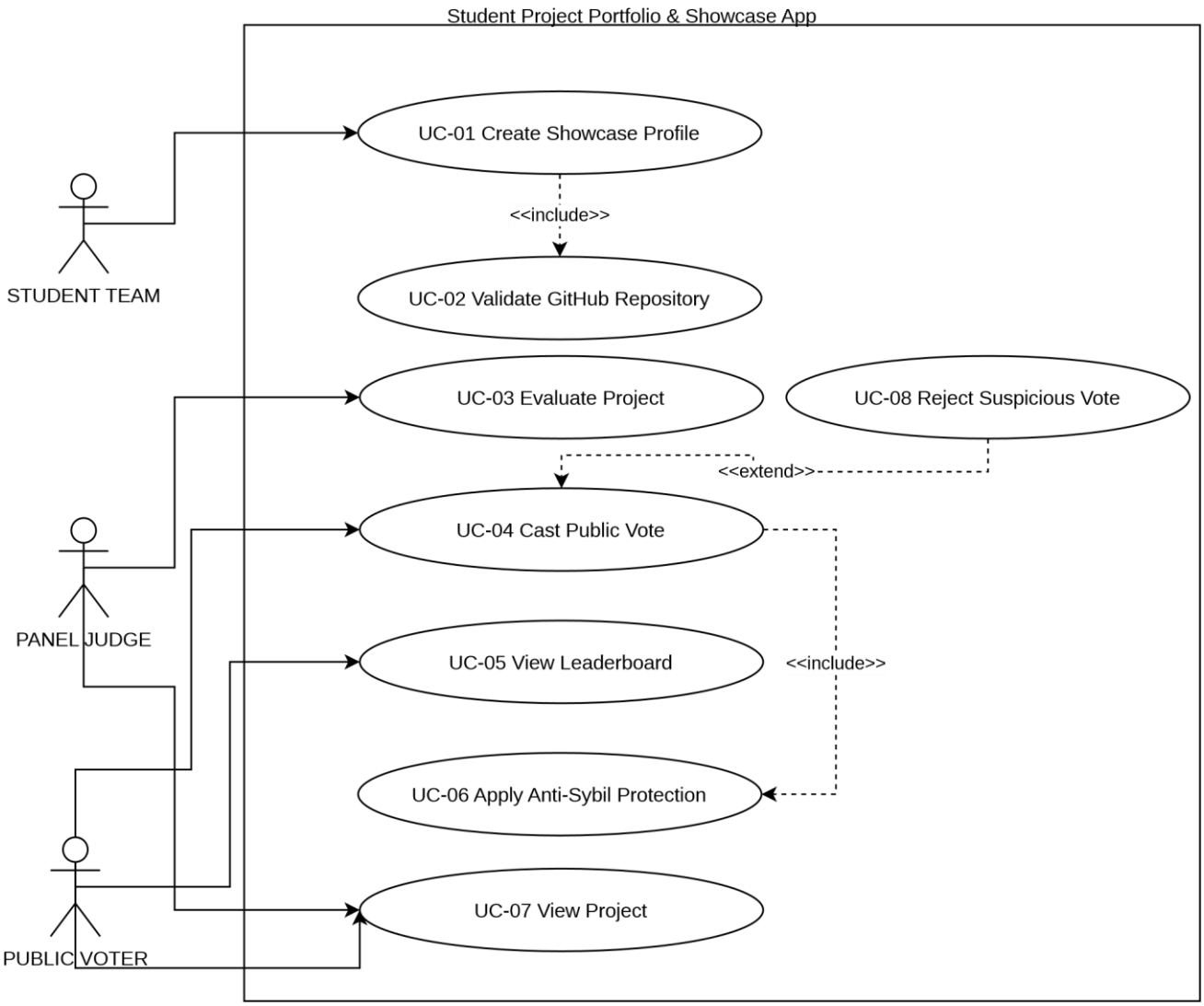
## REQUIREMENTS

## TABLE

| ID | Type | Description | Priority | Acceptance Criteria | Rationale |
|---|---|---|---|---|---|
| FR-001 | Functional | The system shall enable student teams to create their own profiles containing abstract summaries, demo video links, and verified GitHub repository URLs. | High | When the student team enters an abstract, demo video link, and VALID GitHub repository URL the system will accept the input. The system saves the details and displays the new project profile. | Core functionality for showcasing projects. |
| FR-002 | Functional | The system shall allow panel judges to view submitted projects and assign grades using digital evaluation rubrics. | High | When a panel judge opens a submitted project, selects scores for the rubric criteria, and selects “Submit Evaluation,” the system saves the grades and shows an evaluation confirmation. | Enables digital project evaluation. |
| FR-003 | Functional | The system shall allow the public to vote for eligible student projects during the project expo. | High | When a public voter selects an eligible project and clicks “Vote,” the system records the vote and displays a confirmation message. | Enables public participation. |
| FR-004 | Functional | The system shall calculate and display project leaderboards based on judge grades and public voting results. | High | When the system updated rankings. judge grades or public votes, the leaderboard recalculates the project rankings and displays the updated order. | Provides project receives |
| FR-005 | Functional | The system shall prevent a public voter from submitting multiple votes for the same project during the voting period. | Medium | When a voter who has already voted for a project selects “Vote” for the same project again, the system rejects the second vote and keeps the project's vote count unchanged. | Supports fair voting. |
| NFR-001 | Performance & Security | The public voting and leaderboard calculations shall update in real-time with an API response latency under 300 ms. | Medium | When a user submits a vote or requests the leaderboard, the system displays the result within 300 milliseconds. | Ensures responsive voting and rankings. |
| NFR-002 | Security | The system shall enforce anti-sybil protections to prevent unauthorized(People or systems that are not a part of the event) or duplicate voting identities from manipulating public voting results. | High | When the system identifies a duplicate or suspicious voting attempt, it rejects the attempt and does not add the vote to the project's total. | Maintains integrity of public voting. |


Student Project Portfolio & Showcase App
UC-01 Create Showcase Profile
<<include>>
UC-02 Validate GitHub Repository
UC-03 Evaluate Project
UC-08 Reject Suspicious Vote
<<extend>>
UC-04 Cast Public Vote
UC-05 View Leaderboard
<<include>>
UC-06 Apply Anti-Sybil Protection
UC-07 View Project
STUDENT TEAM
PANEL JUDGE
PUBLIC VOTER

**Use Case: Cast Public Vote**

**Primary Actor:** Public Voter

**Goal:** The voter casts a valid vote for a student project during the project expo.

**Preconditions**

1. The project has been published in the showcase system.
2. Public voting is currently open.
3. The project is eligible for public voting.
4. The voter is allowed to participate in the voting process.

**Postconditions**

1. The valid vote is recorded.
2. The project's vote count is updated.
3. The voting result can be reflected in the leaderboard.
4. An invalid or duplicate vote is not counted.

**Main Success Scenario**

1. Voter opens the project showcase.
2. System displays the available student projects.
3. Voter views a project.
4. Voter selects **Vote**.
5. System checks the voter's voting eligibility.
6. System performs the required anti-Sybil protection check.
7. System confirms that the vote is valid.
8. The system records the vote.
9. Vote is recorded.
10. Vote count is updated.
11. Leaderboard is updated.

**Alternate Flow — Duplicate/Suspicious Vote**

6a. System detects that the voting attempt is suspicious or represents a duplicate voting identity.

6a1. System rejects the vote.

6a2. System does not increase the project's vote count.

6a3. System informs the voter that the vote could not be accepted.

6a4. The use case ends.